

Advanced Database Administration II

Backup / recovery (case study)

Goals:

The goal of this assignment is to practice the administration of an oracle database; it gives the student an opportunity to use the information in the course on how to serve as a basis for developing his own backup, maintenance, and recovery



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Introduction:

The Oracle Database (commonly referred to as *Oracle RDBMS* or simply as *Oracle*) is a relational database management system (RDBMS) produced and marketed by Oracle Corporation. As of 2010, Oracle has been a major presence in database computing for many years. Larry Ellison and his friends and former co-workers Bob Miner and Ed Oates started the consultancy Software Development Laboratories (SDL) in 1977. SDL developed the original version of the Oracle software. The name *Oracle* comes from the code-name of a CIA-funded project Ellison had worked on while previously employed by Ampex.

Homework text:

Assumptions:

- 1- The target database is running on windows 2000 platform.
- 2- The target database will be Oracle 9i, named **DESTDB**.
- 3- A user is created in the target database named **BKuser** with a password **BKpass**, and this user is granted **SYSDBA** privileges.
- 4- The database is running in **NOARCHIVE** mode.

This case study is divided into the following sections:

- 1- Create an RMAN script for configuring RMAN parameters.
- 2- Backup the target database.
- 3- Validate the database:
 - a. Verify that database is restorable.
 - b. Verify that tablespaces are restorable.
- 4- Run Recovery catalog maintenance against the target database's control file.
- 5- Remove the target database to simulate and setup a disaster recovery situation.
- 6- Restore and recover the target database.

Requests and Answers:

Please answer the following questions with respect to all the previous requirements:

Question 1:

Write the commands that will login to the target database, with no recovery catalog:

```
c:\> set oracle_SID = DESTDB
c:\> Rman target BKuser\BKpass
```

Question 2:

Write the script that will set the default values for the following parameters:

Retention policy	Redundancy 2
Device type	Disk
Control file backup	Automatic
Control file backup destination	D:\Oracle9i\OraBackups\
Degree of parallelism	2

Rman>

- Configure retention policy to redundancy 2;
- Configure default device type to disk;
- Configure controlfile autobackup on;
- Configure controlfile autobackup format for device type disk to 'D:\Oracle9i\OraBackups\';
- Configure device type disk parallelism 2;

C:\WINDOWS\system32\cmd.exe - Rman target BKuser\BKpass

RMAN> Configure retention policy to redundancy 2;

old RMAN configuration parameters:
CONFIGURE RETENTION POLICY TO REDUNDANCY 2;
new RMAN configuration parameters:
CONFIGURE RETENTION POLICY TO REDUNDANCY 2;
new RMAN configuration parameters are successfully stored

RMAN> Configure default device type to disk;

old RMAN configuration parameters:
CONFIGURE DEFAULT DEVICE TYPE TO DISK;
new RMAN configuration parameters:
CONFIGURE DEFAULT DEVICE TYPE TO DISK;
new RMAN configuration parameters are successfully stored

RMAN> Configure controlfile autobackup on;

old RMAN configuration parameters:
CONFIGURE CONTROLFILE AUTOBACKUP ON;
new RMAN configuration parameters:
CONFIGURE CONTROLFILE AUTOBACKUP ON;
new RMAN configuration parameters are successfully stored

RMAN> Configure controlfile autobackup format for device type disk to ' D:\Oracle9i\OraBackups\';

RMAN-00571: =====
RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====
RMAN-00571: =====
RMAN-03002: failure of configure command at 04/25/2011 23:05:40
RMAN-06492: controlfile autobackup format " D:\Oracle9i\OraBackups\" must specify a "%F" format specifier

RMAN> Configure device type disk parallelism 2;

old RMAN configuration parameters:
CONFIGURE DEVICE TYPE DISK PARALLELISM 2;
new RMAN configuration parameters:
CONFIGURE DEVICE TYPE DISK PARALLELISM 2;
new RMAN configuration parameters are successfully stored

RMAN> _

Question 3:

What Keep in mind that the only backups that are allowed are:

- 1- Whole database backups when the database is cleanly closed and the instance is mounted.
- 2- Tablespace backups of tablespaces that are offline clean or read only.

Write the script that will back up the whole database, after taking the two previous points in consideration.

```
RMAN>
startup force dba;
shutdown immediate;      #to close the database consistently
startup mount;
backup database;
alter database open;
```

Question 4:

Write a script to verify that the entire database can be restored using the existing backups, and check on the ability to restore an individual tablespace named redo_tbs.

- Verify that a the entire database can be restored using the existing backups:

```
C:\> rman target BKuser\BKpass@ DESTDB
```

```
RMAN> run { restore database validate;}
```

- Ability to restore an individual tablespace:

```
RMAN> run { restore tablespace read_only_tbs validate;}
```

Question 5:**Write a script to do the following:**

1. Display a list of files that need to be backed up based on the retention policy.
2. Delete any un-necessary backups.
3. Get a complete summary list of existing backups.

```

- List of files that need to be backed up
  RMAN> REPORT NEED BACKUP REDUNDANCY 2;

- Delete any un-necessary backups.
  RMAN> DELETE OBSOLETE;

- Get a complete summary list of existing backups.
  RMAN> LIST BACKUP SUMMARY;

```

Question 6:

We removed the target database to simulate and setup a disaster recovery situation, the instance is shutdown, and the entire configuration files are in their appropriate places, write a script to restore and recover the target database.

```

C:\ Rman target BKuser\BKpass

RMAN> startup nomount;
RMAN> Allocate channel destdb type disk;
RMAN> restore controlfile from autobackup;
RMAN> alter database mount;
RMAN> restore database;
RMAN> recover database noredo;
RMAN> alter database open resetlogs;

```

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